

E-COMMERCE SALES

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PROBLEM BACKGROUND

Olist is a Brazilian e-commerce platform. The raw transaction data contains missing values and requires preprocessing before analysis. The objective is to identify the main factors affecting customer satisfaction and logistics performance.



DATASET

Brazilian E-Commerce Public Dataset by Olist

The dataset consists of 118,000 records and 39 columns obtained from merging multiple datasets. It includes customer, order, shipping, and timestamp information throughout the order lifecycle.

Key variables: Order ID, price, shipping cost, customer/seller location, payment method, and review score.

ANALYSIS PROCESS

1. Data Cleaning

- Remove missing value
- Handle outliers

2. EDA

- Visualize data to identify patterns and trends

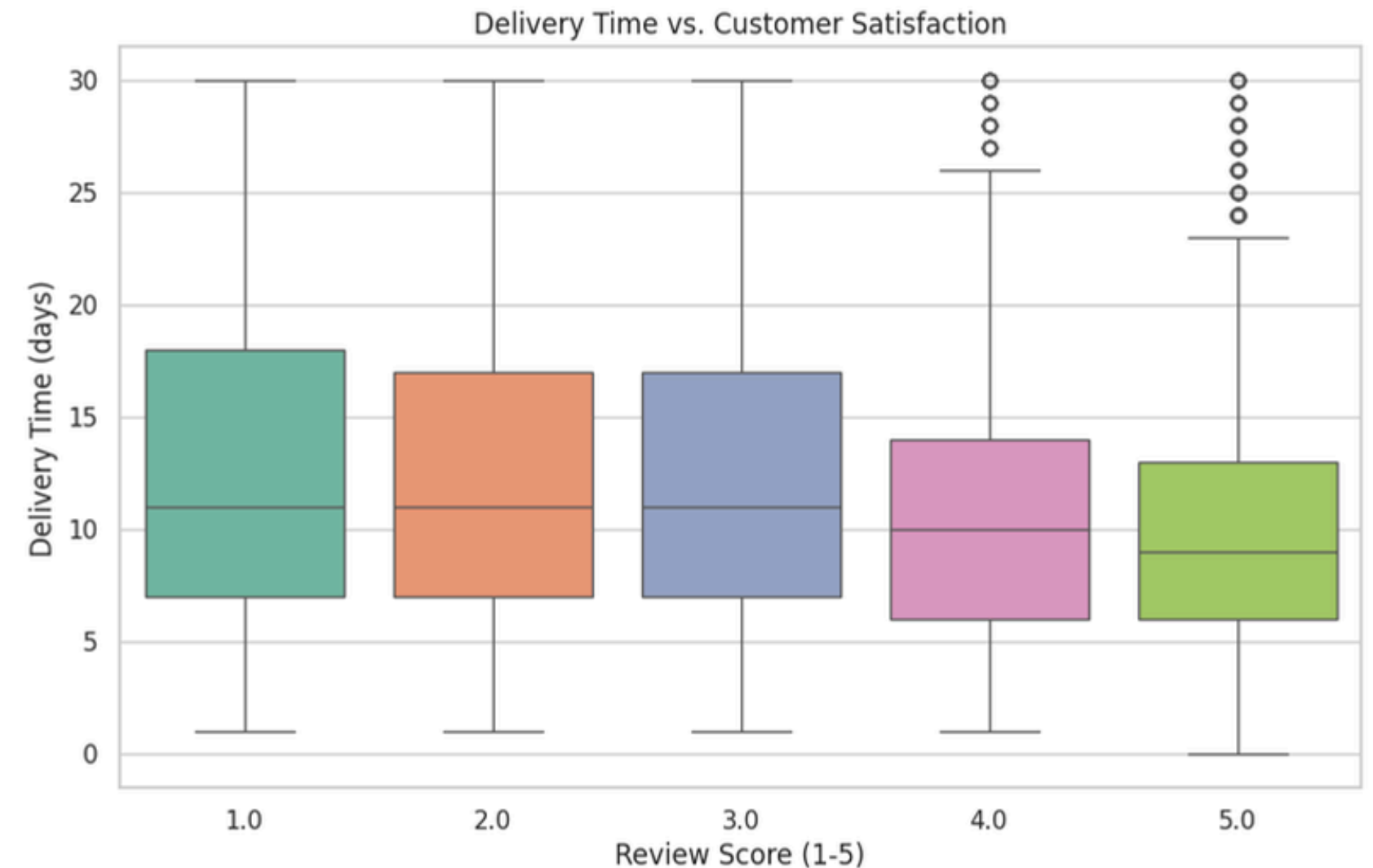
3. Modeling

- Apply machine learning algorithms to support future business decisions

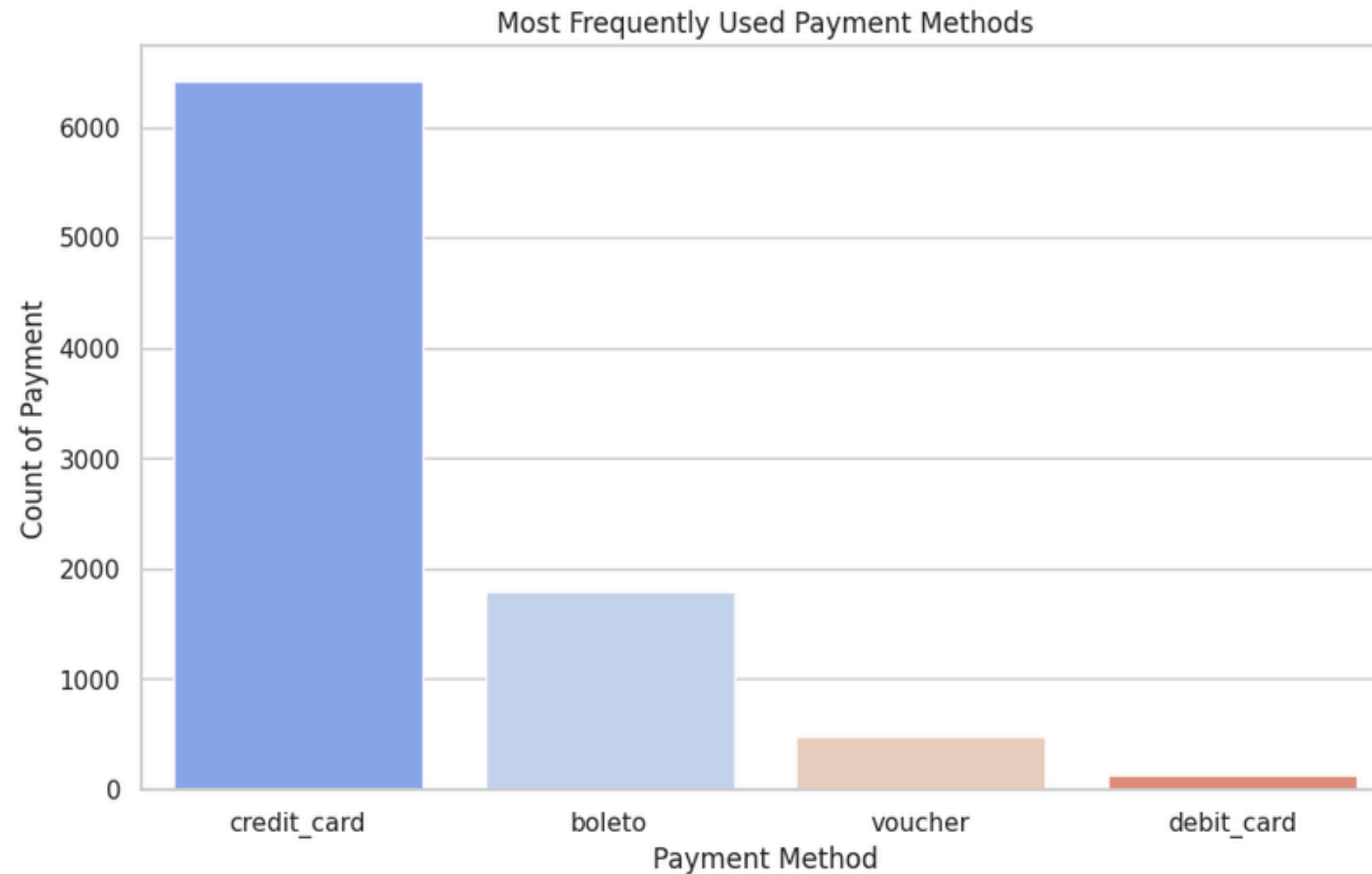
Key Findings #1

Orders delivered within 9-10 days consistently receive high review scores (4-5).

Delivery delays are associated with lower customer ratings and reduced satisfaction.



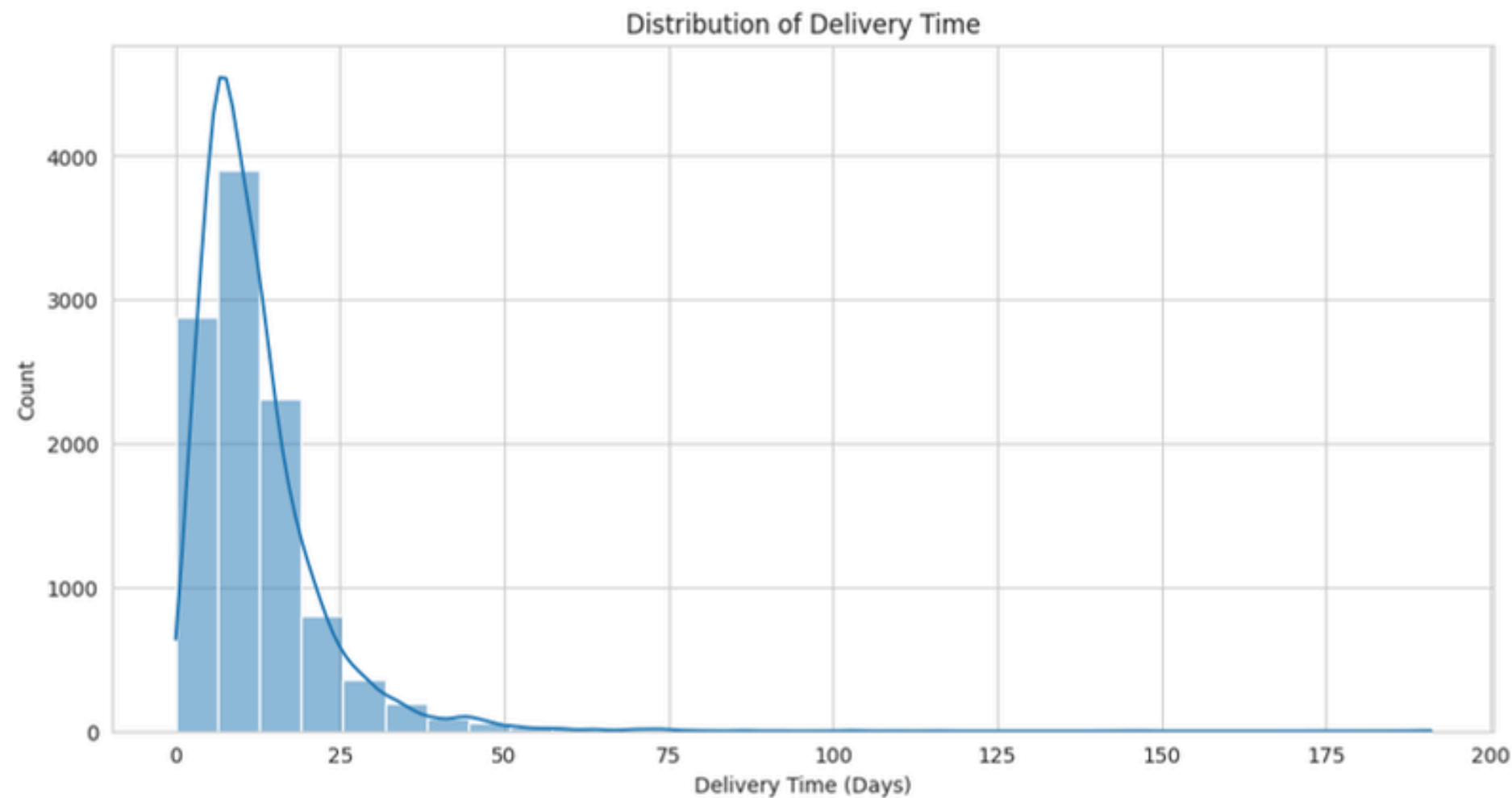
Key Findings #2



The chart shows customer payment method preferences. Credit cards are the most commonly used payment method among customers.

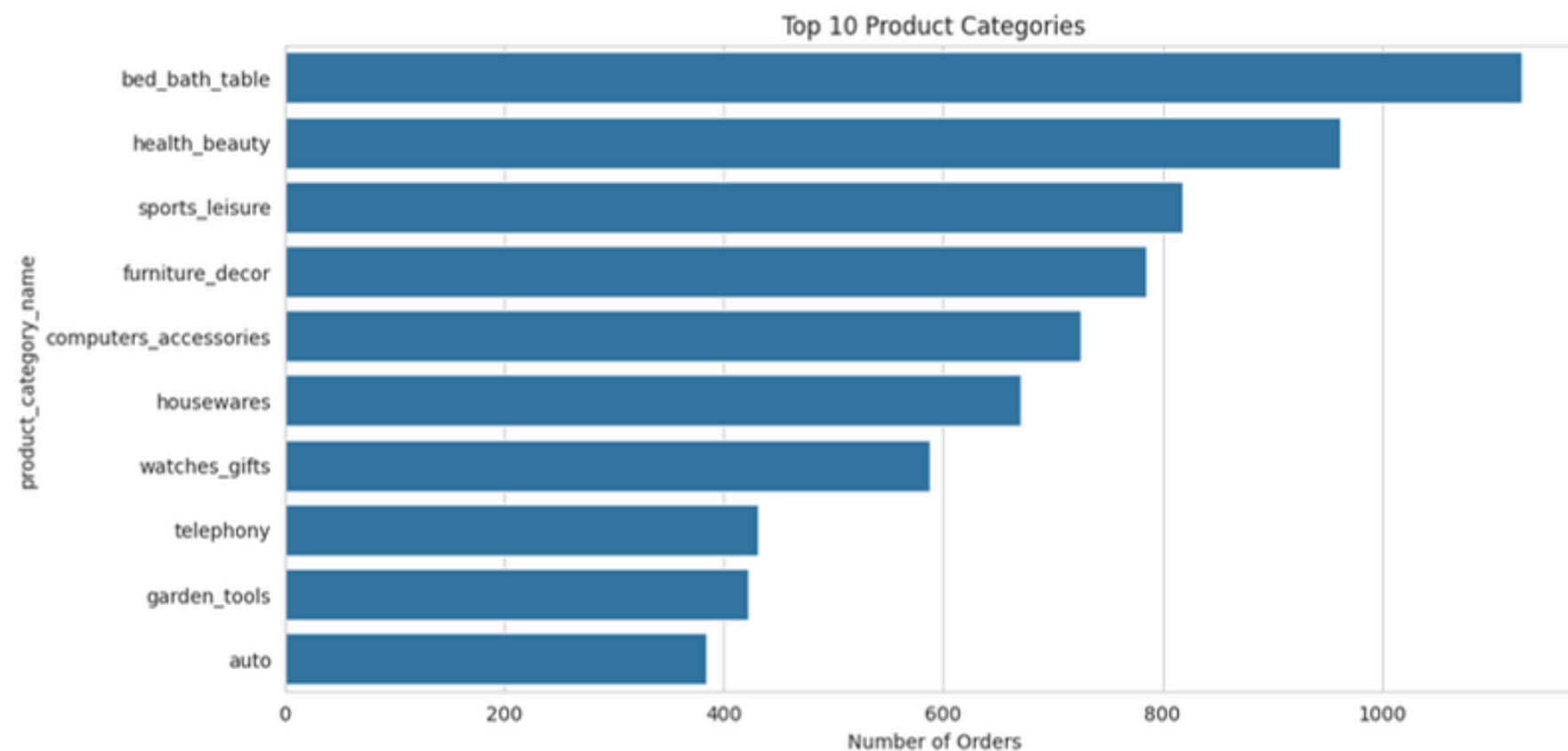
The distribution of delivery time shows that most orders were delivered within a relatively short period. However, there are several outliers with extremely long delivery times, which may negatively affect customer satisfaction.

Key Findings #3

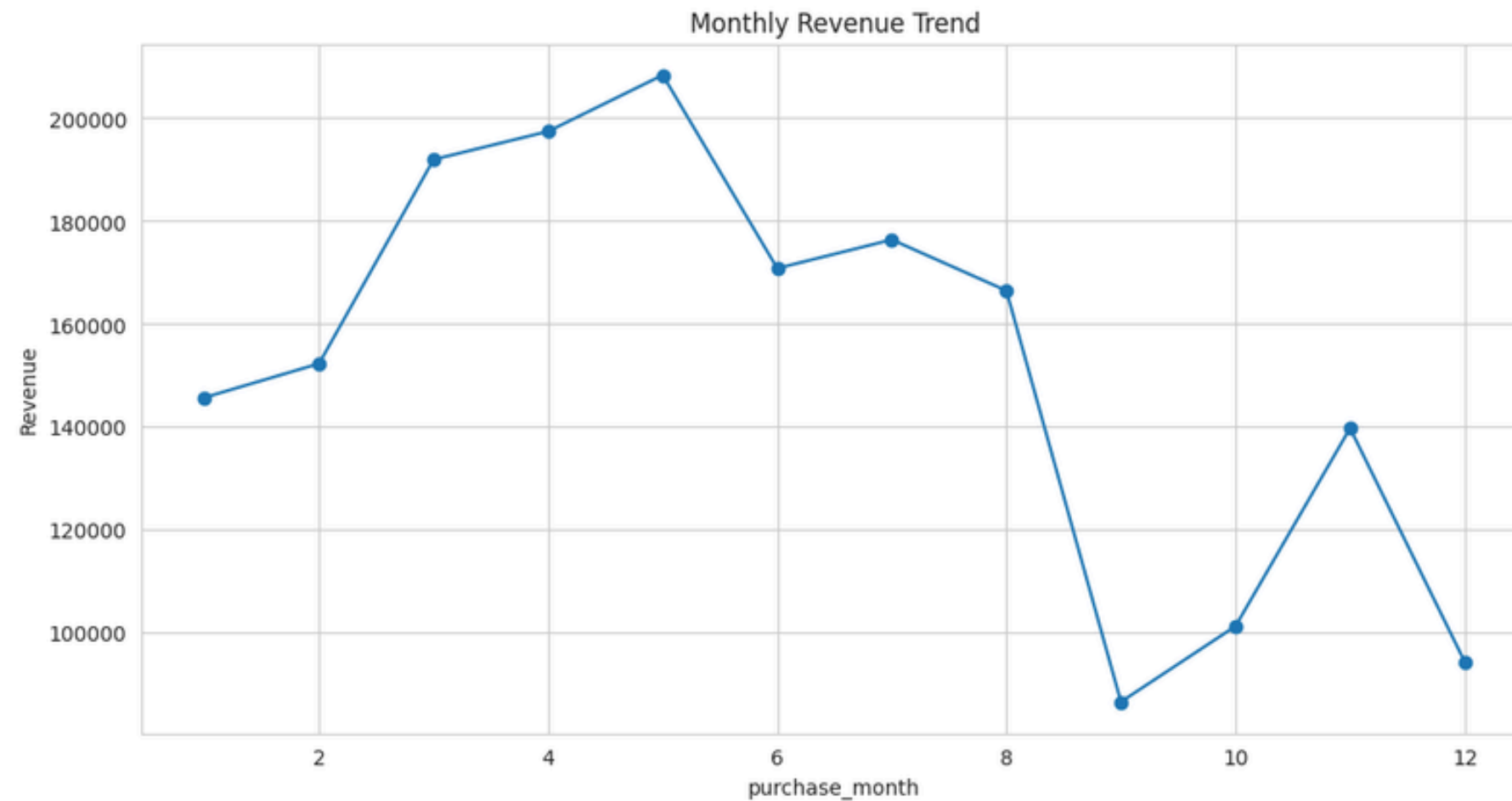


Key Findings #4

Several product categories have significantly higher transaction volumes compared to others. This indicates that customers have stronger preferences for certain product categories on the e-commerce platform.



Key Findings #5



Revenue fluctuated across different months, with some periods showing significant increases. This indicates the presence of seasonal patterns or higher purchasing activity during certain periods.

DATA MODELING

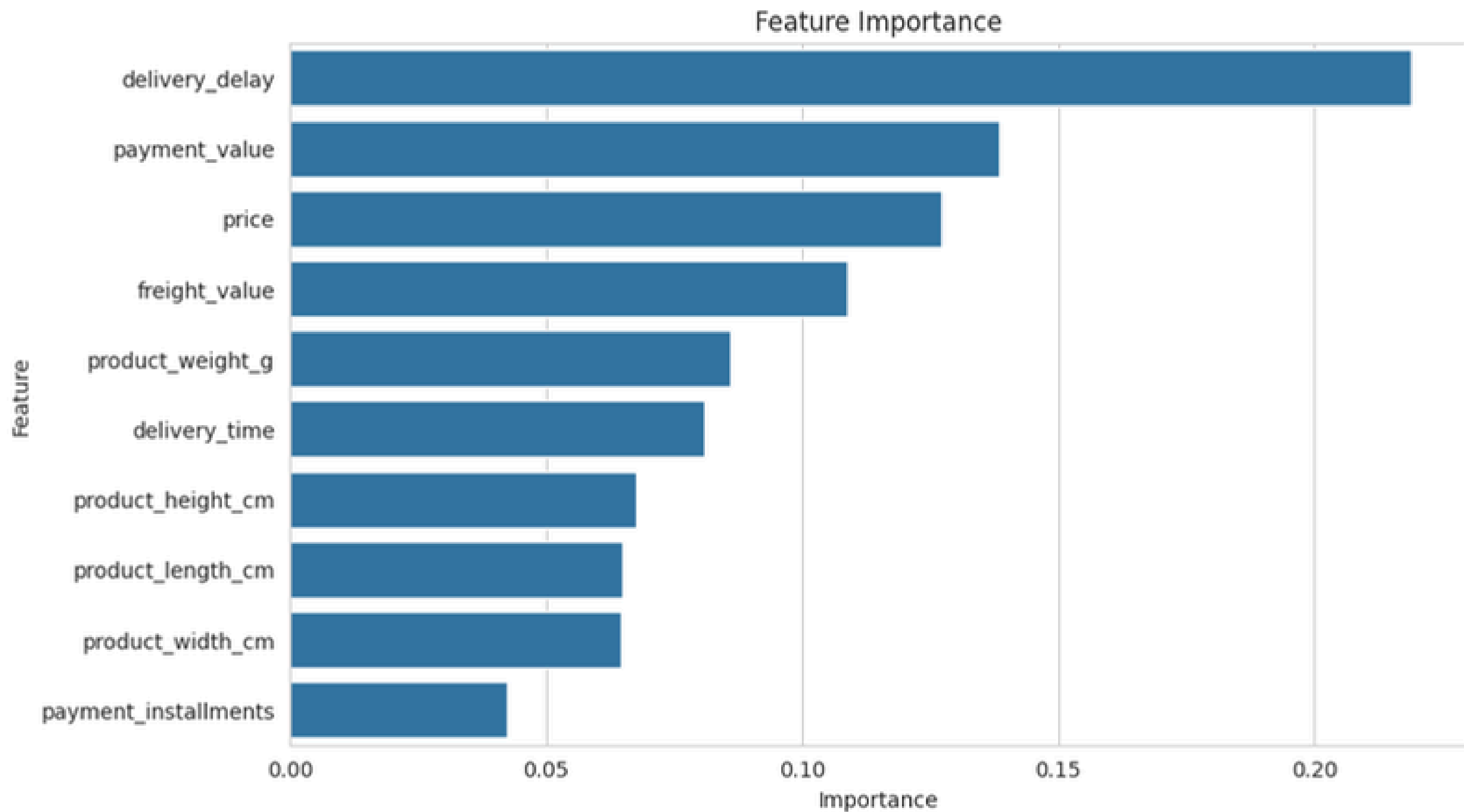
- ALGORITHM: RANDOM FOREST REGRESSION
 - RANDOM FOREST REGRESSION WAS CHOSEN FOR ITS ABILITY TO HANDLE COMPLEX DATA AND PROVIDE RELIABLE PREDICTIONS.
- TARGET VARIABLE: CUSTOMER REVIEW SCORE (REVIEW_SCORE)
 - THE MODEL AIMS TO PREDICT CUSTOMER SATISFACTION BASED ON THE REVIEW SCORES GIVEN AFTER A PURCHASE.

DATA MODELING

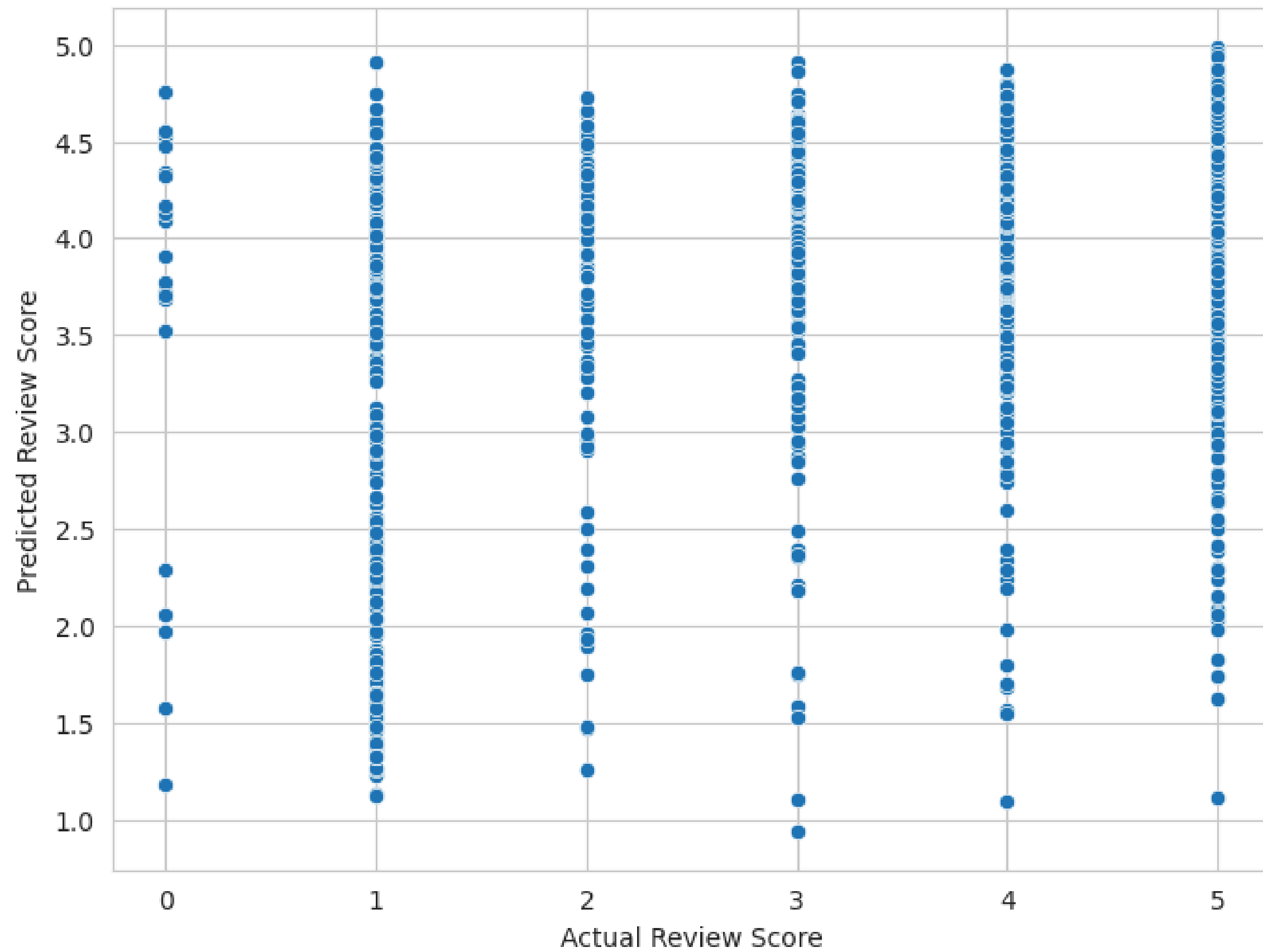
- TRAIN-TEST SPLIT: 80% TRAINING, 20% TESTING
 - THE DATASET WAS DIVIDED INTO TRAINING AND TESTING DATA TO ENSURE THE MODEL COULD BE EVALUATED ON UNSEEN DATA.
- MODEL CONFIGURATION: 100 DECISION TREES
 - THE MODEL WAS TRAINED USING 100 DECISION TREES TO IMPROVE PREDICTION ACCURACY AND REDUCE OVERFITTING.

DATA MODELING

- EVALUATION METRICS: MAE, RMSE, AND R^2 SCORE
 - THESE METRICS WERE USED TO MEASURE PREDICTION ERROR AND EVALUATE HOW WELL THE MODEL EXPLAINS CUSTOMER REVIEW SCORES.
- OUTPUT: CUSTOMER SATISFACTION PREDICTION AND KEY FACTORS ANALYSIS
 - THE MODEL PREDICTS CUSTOMER SATISFACTION AND HIGHLIGHTS THE MAIN DRIVERS OF CUSTOMER REVIEWS.



Actual vs Predicted



BUSINESS RECOMMENDATIONS

1. The company should prioritize delivery optimization because `delivery_time` was identified as the most influential factor affecting customer satisfaction.
2. Product categories with lower review scores should undergo quality evaluation and packaging improvements.
3. Regions with consistently high delivery times may require logistics or warehouse optimization.

THANK YOU